



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

SHORT ANSWER TEST

Code of Conduct:(CSA0716)

I am P.B.VAMSI KRISHNA (192525147) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

P.B.VAMSI KRISHNA

Signature of the student:

Q1. A company needs to transfer a 150 MB design file from a client computer remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

SOL: **Step 1: Number of Segments**

$$\text{Number of Segments} = \frac{150,000,000}{1500} = 100,000$$

Answer: 100,000 segments

Step 2: Number of Frames

Each segment is placed into one frame.

$\text{Number of Frames} = 100,000$

Step 3: Total Data Link Layer Overhead

Overhead per frame = **40 bytes**

$$100,000 \times 40 = 4,000,000 \text{ bytes}$$

Answer:

- **4,000,000 bytes**
- **4 MB** of overhead

Step 4: Total Data to be Transmitted

Original file:

$$150,000,000 \text{ bytes}$$

Add overhead:

$$150,000,000 + 4,000,000 = 154,000,000 \text{ bytes}$$

Convert to bits:

$$154,000,000 \times 8 = 1,232,000,000 \text{ bits}$$

Step 5: Transmission Time

$$\begin{aligned} \text{Time} &= \frac{1,232,000,000}{50,000,000} \\ &= 24.64 \text{ seconds} \end{aligned}$$

Q2 .A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames,calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

SOL: Given:

- **Sliding window size** = 6 frames
- **Data per frame** = 1024 bytes
- **Total frames to transmit** = 48 frames

Step 1: Number of Transmission Windows

$$\text{Number of Windows} = \frac{48}{6} = 8$$

Answer: 8 transmission windows

Step 2: Total Retransmissions

Given:

- **1 frame is lost in every window**
- **Number of windows** = 8

$$\text{Retransmissions} = 8 \times 1 = 8$$

$$=8$$

Q3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the totalheader overhead introduced. Explain how the Network Layer ensures successfuldelivery of fragmented packets to the destination.

SOL: Given:

- **IP packet size = 12,000 bytes**
- **MTU = 1500 bytes**
- **IP header per fragment = 20 bytes**

Step 1: Maximum Data per Fragment

Each fragment must include a **20-byte IP header**.

$$\text{Maximum data per fragment} = 1500 - 20 = 1480 \text{ bytes}$$

Step 2: Number of Fragments

Original data = **12,000 bytes**

$$\text{Number of Fragments} = \left\lceil \frac{12000}{1480} \right\rceil = \lceil 8.108 \rceil = 9$$

Answer: 9 fragments

Step 3: Total Header Overhead

Each fragment has a **20-byte IP header**.

$$9 \times 20 = 180 \text{ bytes}$$

Answer: 180 bytes of header overhead.

Q4 A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

SOL: Given:

- **Sliding window size = 6 frames**
- **Each frame size = 1024 bytes**

- Total frames to transmit = **48 frames**

1. Number of transmission windows required

$$\begin{aligned}\text{Number of Windows} &= \frac{\text{Total Frames}}{\text{Window Size}} \\ &= \frac{48}{6} = 8\end{aligned}$$

Answer: 8 transmission windows

2. Total number of retransmissions

Given that **one frame is lost in every window** and must be retransmitted.

- Number of windows = **8**
- Lost frames per window = **1**

$$\text{Total Retransmissions} = 8 \times 1 = 8$$

Q5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

SOL: Given:

- Video file size = **600 MB**
- Synchronization checkpoint after every **60 MB**
- Failure occurs after **390 MB**

1. Number of checkpoints created before failure

Checkpoints are inserted every **60 MB**.

Checkpoints are at:

- 60 MB
- 120 MB
- 180 MB

- 240 MB
- 300 MB
- 360 MB

The next checkpoint (420 MB) is **not reached** because the failure occurs at **390 MB**.

$$\text{Number of checkpoints} = \frac{360}{60} = 6$$

Answer: 6 checkpoints

2. Data to be retransmitted after recovery

The last successful checkpoint is at **360 MB**.

Failure occurs at **390 MB**.

Therefore, data transmitted after the last checkpoint is:

$$390 - 360 = 30 \text{ MB}$$

After recovery, transmission resumes from the **360 MB checkpoint**.

Answer: 30 MB must be retransmitted.

Q6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

SOL: Given:

- Original file size = **900 MB**
- Compression = **40%**
- Encryption overhead = **5% of the compressed size**

1. Calculate the compressed file size

A **40% compression** means the file size is reduced by 40%.

$$\begin{aligned} \text{Compressed Size} &= 900 \times (1 - 0.40) \\ &= 900 \times 0.60 = 540 \text{ MB} \end{aligned}$$

Answer: 540 MB

2. Calculate the encrypted file size

Encryption adds **5% overhead** to the compressed size.

$$\text{Encryption Overhead} = 540 \times 0.05 = 27 \text{ MB}$$

$$\text{Encrypted File Size} = 540 + 27 = 567 \text{ MB}$$

Answer: 567 MB

3. Calculate the total percentage reduction

Reduction compared to the original file:

$$900 - 567 = 333 \text{ MB}$$

Percentage reduction:

$$\frac{333}{900} \times 100 = 37\%$$

Answer: 37% reduction

Q7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

SOL: Given:

- Total data = **3 GB**
- Throughput = **120 Mbps**
- Request size = **3 MB**

Assume **1 GB = 1024 MB**.

1. Calculate the total number of requests

Convert 3 GB into MB:

$$3 \times 1024 = 3072 \text{ MB}$$

Each request carries **3 MB**.

$$\text{Number of Requests} = \frac{3072}{3} = 1024$$

Answer: 1024 requests

2. Calculate the total transmission time

Convert total data to megabits (Mb):

$$3072 \text{ MB} \times 8 = 24576 \text{ Mb}$$

Transmission time:

$$\begin{aligned} \text{Time} &= \frac{\text{Total Data (Mb)}}{\text{Throughput (Mbps)}} \\ &= \frac{24576}{120} = 204.8 \text{ seconds} \end{aligned}$$

Answer: 204.8 seconds (approximately 205 seconds)

Q8. A packet experiences the following processing delays at different OSI layers:

Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data

Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms,

and the transmission delay is 12 ms. Calculate the total end-to-end

experienced by the packet and explain how each layer contributes to communication process.

SOL: Given:

- Application Layer delay = **4 ms**
- Transport Layer delay = **3 ms**
- Network Layer delay = **5 ms**
- Data Link Layer delay = **2 ms**
- Physical Layer delay = **1 ms**
- Propagation delay = **18 ms**

- Transmission delay = **12 ms**

1. Calculate the total processing delay

$$\text{Processing Delay} = 4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

2. Calculate the total end-to-end delay

$$\begin{aligned} \text{Total Delay} &= \text{Processing Delay} + \text{Transmission Delay} + \text{Propagation Delay} \\ &= 15 + 12 + 18 = 45 \text{ ms} \end{aligned}$$

Answer: 45 ms

Q9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

SOL: Given:

- Useful data = **80 MB**
- Frame size = **1600 bytes**
- Protocol overhead per frame = **80 bytes**

Step 1: Calculate useful data per frame

Useful data in each frame:

$$= 1600 - 80 = \mathbf{1520 \text{ bytes}}$$

Step 2: Convert 80 MB to bytes

Using 1 MB = 1024 × 1024 bytes:

$$80 \text{ MB} = 80 \times 1024 \times 1024 = \mathbf{83,886,080 \text{ bytes}}$$

Step 3: Calculate the number of frames

$$\text{Number of frames} = \frac{83,886,080}{1520} = 55,188.21$$

Since a partial frame cannot be sent, round up:

Number of frames = 55,189

Step 4: Calculate total overhead

$$\text{Total overhead} = 55,189 \times 80 = 4,415,120 \text{ bytes}$$

≈ 4.21 MB

Step 5: Calculate transmission efficiency

$$\text{Efficiency} = \frac{\text{Useful Data}}{\text{Total Data Transmitted}} \times 100$$

Total data transmitted:

$$55,189 \times 1600 = 88,302,400 \text{ bytes}$$

Therefore,

$$\text{Efficiency} = \frac{83,886,080}{88,302,400} \times 100 = 94.9996\% \approx 95\%$$

Q10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

SOL:

Given

- Original medical record size = **240 MB**
- Presentation Layer compression = **30%**
- Transport Layer segment size = **1400 bytes**
- Data Link Layer overhead per frame = **60 bytes**

Step 1: Calculate the compressed file size

Compression = **30%**

Compressed size = 70% of original size

$$240 \times 0.70 = 168 \text{ MB}$$

Convert to bytes:

$$168 \times 1024 \times 1024 = 176,160,768 \text{ bytes}$$

Compressed file size = 168 MB (176,160,768 bytes)

Step 2: Calculate the total number of segments

Each segment carries **1400 bytes**.

$$\frac{176,160,768}{1400} = 125,829.12$$

Round up to the next whole segment:

Total number of segments = 125,830

Step 3: Calculate the total protocol overhead

Overhead per frame = **60 bytes**

$$125,830 \times 60 = 7,549,800 \text{ bytes}$$

≈ 7.20 MB

Total protocol overhead = 7,549,800 bytes (≈ 7.20 MB)

Step 4: Calculate the final amount of data transmitted

Final data = Compressed data + Overhead

$$176,160,768 + 7,549,800 = 183,710,568 \text{ bytes}$$

Convert to MB: $\frac{183,710,568}{1024 \times 1024} \approx 175.2 \text{ MB}$

Final data transmitted across the Physical Layer = 183,710,568 bytes (≈ 175.2 MB)